

1 Laser Beam Propagation

$$\Lambda_0 = 2 \quad (1)$$

$$\lambda = 1.55 \text{ } \mu\text{m}$$

$$W_0 = 0.03 \text{ m}$$

$$F_0 = 500 \text{ m}$$

$$L = 500 \text{ km}$$

$$\Theta_0 = 1 - \frac{L}{F_0} = 1 - \frac{500 \text{ km}}{500 \text{ m}} = -999$$

$$\Lambda_0 = \frac{2L}{kW_0^2} = \frac{2 \cdot 500 \text{ km}}{\frac{2\pi}{1.55\mu\text{m}} \cdot (0.03 \text{ m})^2} = 274.1$$

$$W = W_0 \sqrt{\Theta_0^2 + \Lambda_0^2} = 0.03 \cdot \sqrt{999^2 + 274.1^2} = 31.08 \text{ m}$$

$$I(0, L) = I_0 \cdot \frac{W_0^2}{W^2} = I_0 \cdot \frac{0.03^2}{31.08^2} = I_0 \cdot 0.931 \frac{\text{mW}}{\text{m}^2}$$

$$P_{R_x} = P_{T_x} \cdot \left(1 - e^{\frac{-2r^2}{w^2}}\right) = 10 \text{ W} \cdot \left(1 - e^{\frac{2 \cdot 0.5^2}{31.08^2}}\right) = 5.17 \text{ mW}$$

$$I_{R_x} = 4 \frac{P_{R_x}}{\pi D_{R_x}^2}$$

Gaussian Optics

$$I(r) = I_0 \left(\frac{\omega_0}{\omega}\right)^2 e^{-2\frac{r^2}{\omega^2}}$$

$$2\omega = \frac{\sqrt{2} \cdot \text{FWHM}}{\sqrt{\ln 2}} = 1.699 \times \text{FWHM}$$

$$I(r) = I_0 \cdot e^{-2r^2/\omega^2}$$

$$0.5 = e^{-2r^2/\omega^2}$$

2 Signal to Noise Ratio

$$C = B \cdot \log_2 \left(1 + \frac{S}{N}\right)$$

$$i_S = \frac{\eta e P_s}{h\nu} = \frac{0.85 \cdot e \cdot 10 \text{ mW}}{h \cdot 193.4 \text{ THz}} = 0.01063$$

$$\langle i_N^2 \rangle = 2eBi_S = 2 \cdot e \cdot 100 \cdot 10^9 \cdot 0.01063 = 3.406 \cdot 10^{-10}$$

$$\text{SNR}_0 = \frac{i_s}{\sigma_N} = \sqrt{\frac{\eta P_S}{2h\nu B}} \rightarrow 10 \cdot \log(\text{SNR}_0) = 74.9 \text{ dB}$$

$$\langle \text{SNR} \rangle = \frac{\text{SNR}_0}{\sqrt{\left(\frac{P_{S0}}{\langle P_S \rangle}\right) + \sigma_I^2(D_G) \text{SNR}_0^2}}$$

$$\langle \text{SNR} \rangle \cong \frac{1}{\sigma_1(D_G)}, \quad \text{SNR}_0 \rightarrow \infty$$

$$Pr_{\text{fade}} = 1 - \frac{1}{2} \int_0^\infty (u) \text{erfc} \left(\frac{\text{TNR} - \langle \text{SNR} \rangle u}{\sqrt{2}} \right) du$$

$$Pr(E) = \langle \text{BER} \rangle = \frac{1}{2} \int_0^\infty p_f(u) \text{erfc} \left(\frac{\langle \text{SNR} \rangle u}{2\sqrt{2}} \right) du$$

3 Gimbal angular speed

$$\tan \theta_z = \frac{v \cdot t}{L} \Rightarrow \theta_z = \arctan \left(\frac{v \cdot t}{L} \right)$$

$$\omega_z = \frac{\partial \theta_z}{\partial t} = \frac{v \cdot L}{(v \cdot t)^2 + L^2}$$

$$\omega_z(x) = \frac{v \cdot L}{x^2 + L}, \quad v \cdot t = x$$

$$\omega_z(\theta_z) = \frac{v}{L} \cos^2 \theta_z$$

4 MTBF

$$\lambda_{\text{system}} = \sum \lambda_i$$

$$\text{MTBF}_{\text{system}} = 1/\lambda_{\text{system}}$$

Fried parameter

$$r_0 = (1.46 C_n^2 k^2 R)^{-3/5} \text{ for Horizontal link}$$

$$r_0 = \left[0.423 k^2 \sec(\theta) \int_{h_0}^H C_n^2(h) dh \right]^{-3/5} \text{ for Vertical link}$$

Greenwood frequency

$$f_g = 2.31 \lambda^{-6/5} \left[\sec(\theta) \int_{h_0}^H C_n^2(h) V_{\text{wind}}^{5/3}(h) dh \right]^{3/5}$$